

Stocker — an AI Investment Team on n8n

Final project · Track 1 — Financial Automation: a Multi-Agent n8n System

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▶ Watch the 5-minute demo

6 n8n workflows 159 nodes	4 specialist agents 3 run in parallel	3 Risk Manager passes draft → critique → final	112 offline tests green in CI	\$0.02–0.07 per run logged per agent
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1 · The financial problem

Forming a view on a stock means synthesizing three unlike sources: price action, news flow, and exchange disclosures. Doing that by hand across a watchlist is slow and inconsistent. Doing it with a single LLM prompt is fast but untrustworthy — one-shot pipelines invent financial figures, hide their uncertainty, and overcommit on thin evidence.

Tel-Aviv names add specific friction. Coverage splits across Hebrew and English, and TASE disclosures live on a bot-protected JavaScript site whose actual figures sit in PDF attachments two layers below the page a naive scraper would read.

What this project solves. Given a watchlist that may mix TA-35 and S&P 500 names, produce a justified **long / short / hold / avoid** recommendation per ticker — with a conviction level, a complete reasoning trace, and citations back to every source. Two guarantees hold throughout: **no financial figure is ever fabricated**, and **no uncertainty is ever hidden**. A failed data source lowers conviction; it never becomes a signal.

2 · What was built

A virtual investment team of four AI agents orchestrated by **n8n**, backed by a local **FastAPI** service that owns all real computation. Three specialists gather independent evidence per ticker; a Risk Manager weighs it through a three-pass self-critique and issues the call. Every run produces a **PDF report** with an executive summary, per-ticker pages, and the full reasoning trace.

Agent	Role	Key mechanism
Sentiment claude-haiku-4.5	Reads recent news from NewsAPI and RSS, English and Hebrew	Dual scoring. An LLM and a fine-tuned transformer (FinBERT for English, DictaBERT for Hebrew) score every article independently. Their disagreement is itself a reported signal.
Earnings grok-4.3	Reads exchange disclosures — Maya for TASE, SEC EDGAR for US, routed by market	Self-consistency. Every figure is sampled three times at temperature 0.3 and committed only on a majority

		vote of verbatim matches. Anything else is marked ambiguous .
Technical gemini-2.5-flash-lite	Narrates RSI, MACD, Bollinger and ATR computed server-side	The model narrates pre-computed JSON only. No number in the report originates inside a language model.
Risk Manager claude-haiku-4.5, ×3	Synthesizes the three signals into the final call	Draft → devil's-advocate critique → final , then a deterministic rubric clamp so a model that breaks the decision rules cannot reach the report.

Three entry points share one pipeline — no duplicated code path:

Entry point	Behaviour
Manual trigger	The whole watchlist on demand, from the n8n editor. Never gated by market hours.
Hourly schedule	A per-market gate analyzes only tickers whose market is open: 11:00 Israel time runs the TASE names, 20:00 runs the US ones — each market carrying its own calendar, hours, news feeds, earnings source and currency.
Chat assistant	<i>"what do you think about Teva?"</i> runs the real pipeline for TEVA.TA in about a minute. Deliberately a router, not an analyst — five nodes and a single tool, so it can only invoke the pipeline and relay what comes back.

3 · Architecture

Two layers joined by one HTTP boundary. n8n holds the orchestration and every LLM call, and contains no machine-learning code. The FastAPI service owns everything needing a real library — indicators (**pandas-ta**), transformer inference, PDF rendering (WeasyPrint over Jinja2), and scraping (Playwright for Maya's protected SPA, plain HTTP for EDGAR) — persisting to **DuckDB**: prices, news, earnings, runs, recommendations, costs.

Each run opens with a run id and the watchlist, fans the three analysis agents out per ticker at concurrency 3, runs the Risk Manager's three passes per ticker, harvests real token usage from n8n's execution API, renders the PDF, and closes the run.

- **Heavy data never crosses the LLM boundary.** Price arrays, article bodies and rendered HTML stay server-side; models see short text and scores.
- **Every LLM response is schema-validated.** A malformed reply earns one stricter retry, then an explicit degraded result — never a silent fallback.
- **Failures degrade, they never crash.** Yahoo, NewsAPI, RSS, Maya, EDGAR and the n8n API each fail into a labelled state that the Risk Manager turns into lower conviction.
- **Costs are observable.** Every call is logged with tokens, dollars and latency; a five-ticker run costs roughly **\$0.04–0.08**.
- Times are stored in UTC and rendered in Asia/Jerusalem. Secrets live only in a gitignored **.env**.

4 · The AI, beyond a baseline pipeline

These are not five separate tricks. They are one principle applied at three scales: **an LLM cannot be trusted to report its own uncertainty, so uncertainty is derived from disagreement between**

independent attempts — never from the model's stated confidence, which HeBERT was willing to put at 0.998 while being wrong (§5.1).

Where disagreement is measured	What disagrees	What it produces
Within one model	three samples of the same extraction prompt, temperature 0.3	a figure that does not survive a majority becomes ambiguous — never a number
Between two models	an LLM against a fine-tuned transformer, same headline	a split above 0.3 caps conviction at medium; both scores are printed
Between two passes	a draft against a devil's advocate written to attack it	objections that survive downgrade the call; all three passes are printed

The system never asks a model “how sure are you?”. Confidence is earned by surviving an independent challenge, and §5.2 measures whether each challenge actually has teeth.

Technique	What it adds
Dual sentiment LLM alongside FinBERT / DictaBERT	An independent, non-LLM opinion on every article. Disagreement above a threshold caps conviction at medium and is printed in the report — model conflict becomes information rather than something averaged away.
Self-consistency extraction n = 3, temperature 0.3, majority vote	“Never invent numbers” enforced by construction rather than by instruction. A figure is committed only when repeated samples agree on text present verbatim in the source; everything else prints as ambiguous , visually distinct in the PDF. Sampling makes it stochastic, so it sharply reduces invented figures and makes the residue <i>measurable</i> — see §5.1.
Three-pass critique loop with a deterministic rubric clamp	The recommendation visibly stress-tests itself, and all three passes are printed. A mechanical clamp then enforces the decision rubric — and when it overrides the model, the report says so rather than quietly rewriting the model's words.
Versioned few-shot prompts with an evaluation harness	Prompts are files, not workflow internals: version-controlled, reviewable, and scored against hand-labelled fixtures by a single command.
Degraded-agent epistemics	A failed agent is defined as <i>no information</i> . The prompts state that an empty result caused by a system failure says nothing about the company, and the rubric counts it for neither side while still lowering conviction.

5 · Results

5.1 · Evaluation harness

Scored against hand-labelled fixtures: 30 news items (20 English, 10 Hebrew), 10 disclosures, and 7 chat-router cases.

Agent	Result
Sentiment — LLM	accuracy 0.90 (27 / 30) · MAE 0.12
Sentiment — transformers, overall	accuracy 0.77 (23 / 30) · MAE 0.39

└ English — FinBERT	accuracy 0.80 (16 / 20) · MAE 0.43
└ Hebrew — DictaBERT	accuracy 0.70 (7 / 10) · MAE 0.32
Sentiment — agreement	Pearson r 0.82 (n = 30)
Earnings — classifier	macro-F1 1.00 · materiality accuracy 0.90 (n = 10)
Earnings — extractor	precision 1.00 · recall 1.00 · ambiguous-when-absent 22 / 22
Chat router	refusal 4 / 4 · routing 3 / 3 · no fabrication

The transformer arm is reported per language, because it is two models and the mean of the two describes neither. Reaching 0.70 on Hebrew meant replacing the model — and the investigation is recorded because two of its steps failed:

Step	Result
HeBERT — the original choice	0.30 . Not weakly discriminating but <i>not discriminating at all</i> : neutral for 10 of 10 Hebrew items at 0.833–0.998 confidence, including “record quarterly orders, guidance raised” at 0.998.
Re-normalise polarity over the polar classes	Worse — 0.53 . The residual mass is noise, not signal.
Threshold sweep, $\pm 0.1 \rightarrow \pm 0.6$	No movement beyond a single item.
DictaBERT — adopted	0.70 . Every negative and every neutral read correctly; it misses only positives.

Agreement between the two arms rose **0.70** → **0.82** as a direct consequence — a model outputting ~0 for everything cannot correlate with anything, so the earlier figure was largely measuring noise.

The extractor row is the “never invent numbers” guarantee, measured — including its limit. Because self-consistency *samples* (three times at temperature 0.3), it is not deterministic: four measured runs scored **ambiguous-when-absent** at 22/22, 21/22, 22/22 and 22/22 — the single miss being a figure committed that its source does not state. The honest claim is that the mechanism sharply reduces invented figures and makes the residue measurable — not that it eliminates them. A single-sample extractor offers neither guarantee nor a way to count its own failures.

These figures **reproduce**. Re-running the harness eleven days later returned every metric identical — accuracy, MAE, Pearson r , macro-F1, precision and recall, and the router's 4/4 and 3/3. Only the cost moved, by a 676-token difference in the earnings arm. That is the intended consequence of pinned fixtures and **temperature = 0** everywhere outside the self-consistency sampler: the one number that *should* wander is the extractor's, and it is reported rather than smoothed.

5.2 · Ablations — do the techniques earn their cost?

Each technique above costs real tokens: a three-sample vote triples the extraction cost, the critique loop triples the Risk Manager's, and a few-shot block inflates every prompt. “We used self-consistency” is a claim about effort, not effect. A second harness (**npm run ablations**) switches each technique off and re-scores.

Technique removed	Measured consequence
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Self-consistency n = 3 vote → n = 1 sample	Invented figures 0 → 2 ; invented rate 0% → 3% ; precision 1.00 → 0.92 . Per individual draw the rate was 5%, 0%, 5% — two of three samples each committed a figure absent from the source, and the vote caught both because the samples disagreed with one another . Disagreement between draws <i>is</i> the detector; a single sample has no access to it.
Few-shot examples 9 examples → none	Sentiment accuracy 0.90 → 0.77 , MAE 0.12 → 0.20 . The 0.77 is the point: it is exactly what the <i>free</i> local transformer arm scores. Without the examples the paid model stops being worth paying for — the few-shot block is the entire margin it contributes over the model it exists to disagree with.
The critique loop final → draft	Over 26 persisted recommendations: the call changed 8 times (31%), <i>every one long</i> → <i>hold</i> ; conviction changed 9 times (35%), 9 downgrades and 0 upgrades . Across every run on record the critique has never once made the system more bullish or more confident. Nor is it a rubber stamp: in 19 of 26 cases the devil's advocate argued for a different call.

Two notes on method. The self-consistency arms are scored from the *same* three draws, so the comparison is paired, immune to sampling luck, and costs no more than one extraction run. The critique arm needs no LLM calls at all — the *draft* is already the counterfactual, recorded before the critique existed to challenge it — so that arm is free and reproducible without an API key.

What these numbers do not show. The samples are small (10 disclosures, 30 news items, 26 recommendations from an overlapping ticker set). The critique arm is observational rather than controlled: its draft was written by a prompt that *knows* a critique is coming, so it is not a clean single-pass baseline. And “changed the call” is not “improved the call” — nothing here measures whether the *hold* was right, which is precisely what the missing backtest (\$7) would test.

5.3 · A silent data-integrity bug, and what finding it changed

The most consequential defect in this project produced no error, no degraded status and no failing test. It was found by reading */ohlcv* output and noticing that **every Sunday bar was an exact copy of the previous Thursday's**.

The Tel Aviv Stock Exchange trades Sunday to Thursday, so the cleaning step reindexed each series onto that week — a rule taken from configuration. Checked against the live feed, **Yahoo Finance returns .TA daily bars on a Monday–Friday index with no Sunday sessions at all** (verified on *TEVA.TA*, *ICL.TA* and *POLI.TA*, all timestamped *Asia/Jerusalem*). Reindexing therefore did two things at once: it **discarded every real Friday bar**, and it **forward-filled a synthetic Sunday** from the preceding Thursday.

Consequence on <i>TEVA.TA</i> , 180 days	Measured
Fabricated bars	35 of 177 rows — 34 of them Sundays
Share of the series that was synthetic	≈ 19%
Real trading sessions discarded	31 Fridays
Effect on indicators	each phantom carries a zero close-to-close return and a near-zero true range, which deflates ATR, flattens RSI and distorts the 8×-MAD outlier test

After the fix

0 fabricated bars; all 31 Fridays retained; 7 genuine mid-week holidays bridged. AAPL byte-identical — the US path was never affected

The fix removes the assumption rather than correcting it. The session grid is now derived from the weekdays the source actually delivers, so no bar can be created on a weekday the feed never sends and no fetched row can be reindexed away — for any market or any future data source. The configured trading week is still compared against what arrived, and a mismatch is *reported* per symbol rather than silently absorbed.

Two things about this are worth more than the fix itself. First, it contradicted the system's own central rule: a pipeline built to refuse inventing an earnings figure was quietly inventing a fifth of its price history. Guardrails hold where they are enforced, not where they are stated — the earnings path had a mechanism, the data path had only an assumption. Second, **the bug outlived its own repair**: the cache wrote with insert-or-replace, which can correct a row and add a row but never *retract* one, so the already-stored phantoms kept feeding the endpoints after the grid was fixed. A re-ingest is now authoritative for the window it covers. Both behaviours are pinned by tests that fail if either returns.

What is still unresolved, and deliberately so

Whether those Friday .TA bars are genuine sessions, sessions labelled a day late, or a feed artefact is **not known** — their volume runs consistently below the Monday–Thursday bars. Rather than guess, the code now makes a narrower claim that holds under every reading: it invents nothing and it discards nothing. An honest open question is worth more than a confident wrong answer, which is the same principle the rest of the system applies to the LLM.

5.4 · Live behaviour, including under failure

A single-ticker chat run completes in about a minute for two to four cents, producing a four-page PDF and complete database rows. A mixed watchlist renders one report carrying per-market currencies and a market-grouped summary.

The reference run behind the committed sample is `r_2026-08-16T12:53`: both markets in one pass, driven through the chat assistant, **245 seconds** and **\$0.0686** across four agents, all three specialists reporting *ok*. It was verified against the persisted rows rather than the chat reply — one *runs* row (*mode* = *chat*), two *recommendations* rows each carrying draft, critique and final, *costs* rows per agent, and a seven-page PDF containing every block the design requires. On the US side the extractor committed **\$109.4 billion** and **\$2.02** from Apple's 8-K with all three samples agreeing, and refused *guidance* as *ambiguous* because the exhibit does not state it — the never-invent rule doing both of its jobs on live data rather than on a fixture.

Case study — what happens when a data source dies

With a rejected news API key, sentiment degrades to zero articles. In an early run the critique pass argued that the *absence* of news was “itself a material anomaly... both scenarios bearish” — inferring a market signal from an infrastructure failure. Root cause: the agent's failure status was not machine-readable to the prompts. After the fix the final call states that “*a degraded agent is neutral by definition... its absence is not a bearish or bullish signal*”, while still lowering conviction. Failure, diagnosis and before/after evidence are in the history.

5.5 · The reasoning trace, as printed in the report

Excerpts from one run's three passes for **NFLX**, after the fix described above. Each pass is printed in full on the ticker's page.

1 · DRAFT	hold · low
"Agreement count is 1 of 3... Per rule 2, one or fewer agents agreeing mandates a 'hold' call regardless of conviction cap."	
2 · DEVIL'S-ADVOCATE CRITIQUE	counter: hold
"MACD below signal line — a subtle bullish inflection the draft dismisses as neutral... the denominator includes a degraded agent that contributed zero information. The system correctly avoids inventing sentiment signal , but the true agreement ratio is understated."	
3 · FINAL DECISION	hold · low
"The critique's observations do not override the mechanical agreement rule... per the rubric, a degraded agent is neutral by definition and does not constitute evidence of direction — its absence is not a bearish or bullish signal. "	

The critique attacks the draft on its own evidence; the final answers each objection explicitly rather than restating the draft. Full reports are committed under [docs/samples/](#).

6 · Risks and caveats

- **The critique loop reduces overconfidence; it does not guarantee correctness.** It is a structured reasoning aid, not a financial-validity guarantee — and the report's footer says so.
- **No live execution and no return measurement.** The system argues positions; it never checks whether they would have made money.
- **Maya scraping is best-effort.** A protected JavaScript site rendered headless, with figures two layers down in a PDF attachment. A scanned file or dead link yields **ambiguous** figures — never guessed ones.
- **EDGAR extraction is a bounded excerpt, not a structured parse.** A 10-Q's tables flatten badly into text, so the selection logic explicitly prefers press-release exhibits.
- **News coverage is uneven.** Measured live: **TEVA.TA** returns 0 items in the default 3-day window against 47–49 for large US names. Auto-derived search terms also fail where a registered name differs from the press name ("Alphabet" vs "Google", hand-curated).
- **A model swap is not a no-op.** Models clear schema validation at different rates and the retry path is a different branch — swapping the earnings model surfaced a latent fan-in bug the previous one had masked.
- **One open item.** The agreement rule counts a degraded agent in its three-agent denominator, so "1 of 3" may really be "1 of 2 measurable" — raised by the system's own critique pass during a live run.
- **Yahoo's Tel Aviv calendar does not match the real trading week, and the reason is unknown.** It returns **.TA** bars Monday–Friday for an exchange that trades Sunday–Thursday (\$5.3). Ingestion now follows the feed's own calendar and reports the mismatch, so it invents no bar and discards none — but whether those Friday bars are real sessions, mislabelled ones, or an artefact is unresolved.

Any TA-35 report generated before this fix understates its indicators; the committed sample was regenerated afterwards.

- **A guardrail is only as good as its enforcement point.** The never-invent rule was mechanised for earnings figures and merely assumed for price data — which is exactly where fabrication appeared. The lesson generalises to anything in this system not currently pinned by a test.
- **Sentiment coverage of TA-35 mid-caps is genuinely thin, not broken.** In the reference run `TEVA.TA` returned zero articles in the window with both scores at 0. The report prints the article count and pads nothing, and the critique names the gap as a reason for lower conviction without reading a direction into it — but a dual-sentiment split cannot inform a call that has no news to score.

7 · Future development

- **Backtest the recommendation stream** against realized returns — turning rationale quality into measured alpha, net of costs.
- **Delivery channels.** E-mail and Telegram are each a single additional n8n node; the report step is already structured for them.
- **Scale the watchlist** to the full TA-35 and S&P 500. The per-market architecture supports it today; a paid news tier is the binding constraint.
- **Rubric refinements.** Resolve the degraded-denominator question, and consider a stronger Risk Manager model — a one-field change, with the cost table as arbiter.
- **OCR fallback** for scanned disclosure PDFs, currently an honest `ambiguous`.

Running it. `npm run setup`, add API keys to `.env`, then `npm run dev` to start the quant service, n8n and the chat UI together; the six workflows are imported once through the n8n editor. Quick-start, evaluation reproduction and wiring details are in [README.md](#); the complete technical design in [docs/design.md](#); run walk-throughs in [docs/results.md](#); the ablation method and its caveats in [docs/ablations.md](#); and the demo outline in [docs/demo_script.md](#). This document is itself built from source — `npm run summary` renders `docs/project_summary.html` to the committed PDF, so the two cannot drift.